`lm()` and `summary()` in R — Visual Guide

Example: A manager asks: *does advertising spend predict sales?* Data: 30 stores. **Y** = Sales ($1000s) | **X** = AdSpend ($1000s)

$$Sales_i = b_0 + b_1 \times AdSpend_i + \varepsilon_i$$

```
model ← lm(Sales ~ AdSpend, data=stores)
summary(model)
```

~ means "depends on" | *data = which data frame*

`summary(model)` output

```
Call:  lm(formula = Sales ~ AdSpend, data = stores)

1 Residuals:
    Min   1Q  Median  3Q   Max
  -8.324 -2.147  0.381  2.503  7.196

Coefficients:
              2          3        4       5
            Estimate Std.Error t.value Pr(>|t|)
(Intercept)  12.351   3.214   3.842 0.00064 ***
AdSpend       2.876   0.412   6.981 1.7e-07 ***

Signif.  codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.'  0.1

6 Residual standard error:  3.891 on 28 degrees of freedom
7 Multiple R-squared:  0.6351,  Adjusted R-squared:  0.6221
8 F-statistic:  48.74 on 1 and 28 DF,  p-value:  1.7e-07
```

1 Residuals
Errors = actual − predicted.
Five-number summary of all prediction errors.

2 Estimate
The coefficients b_0, b_1.
These are the numbers in your regression equation.

3 Std. Error
How uncertain is the estimate?
Smaller = more precise.
Larger = less reliable.

4 t value
= Estimate / Std. Error.
How far is b from 0?
Tests H_0: $b=0$ vs H_1: $b\neq 0$

5 p-value
Prob. of this result if $b=0$.
Smaller = more evidence.
Tests H_0: $b=0$ vs H_1: $b\neq 0$

6 Residual SE
Predictions off by ~$3,891.
Same units as Y. Smaller = better.
$df = n-2 = 30-2 = 28$

7 $R^2 = 0.6351$
63.5% of variation explained.
Range: 0 to 1.
Adj. R^2: corrected for # of variables.

8 F-statistic
Tests H_0: *all* slopes = 0.
p: 1.7e-07 → reject H_0.
Same p as t-test when 1 predictor.

Intercept: $b_0 = 12.351$
Sales when ads = $0: **$12,351**.
May not be meaningful if no store spends $0.

Slope: $b_1 = 2.876$
+$1k in ads → **+$2,876** in sales.
This is the key business insight!

`1.7e-07` = ?
$1.7 \times 10^{-7} = 0.00000017$
Move decimal 7 places left.